

torch.optim.Optimizer.zero_grad#

[https://pytorch.org/docs/stable/generated/torch.optim.Optimizer.zero_grad.ht](https://pytorch.org/docs/stable/generated/torch.optim.Optimizer.zero_grad.html)

Description:

Reset the gradients of all optimized `torch.Tensor` s. `set_to_none` (bool, optional) – Instead of setting to zero, set the grads to `None`. Default: `True`

This will in general have lower memory footprint, and can modestly improve performance. However, it changes certain behaviors. For example: When the user tries to access a gradient and perform manual ops on it, a `None` attribute or a `Tensor` full of 0s will behave differently.

Function Signature

```
Optimizer.zero_grad(set_to_none=True)[source]#
```

Parameters

Detailed Documentation

Documentation

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`set_to_none` (bool, optional) –

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When the user tries to access a gradient and perform manual ops on it, a None attribute or a Tensor full of 0s will behave differently.

If the user requests `zero_grad(set_to_none=True)` followed by a backward pass, `.grads` are guaranteed to be None for params that did not receive a gradient.

`torch.optim` optimizers have a different behavior if the gradient is 0 or None (in one case it does the step with a gradient of 0 and in the other it skips the step altogether).